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OPERATING SYSTEMS AND SYSTEMS PROGRAMMING (25CS2104E)

2026–27, Term-I

Project Problem Statement Submission Form

Section / Team Details

Section No.: 1

Team No.:

Project Title: Linux System Information and Resource Monitoring Tool

Team Members

Roll Number	Student Name	Signature
2520030554	Sai Hima Sekhar	
2520030019	V. Navadeep	
2520030063	K. Shankar	

1. Abstract

(Summarize the project idea, problem background, relevant Operating Systems/System Programming concepts, proposed Linux-based solution, and expected outcome. The abstract should not exceed 250 words.)

The Linux System Information and Resource Monitoring Tool is a user-space system programming application designed to provide users with a clear overview of the system's hardware, operating system, processes, and resource utilization. Monitoring system resources is important for identifying high CPU usage, memory consumption, running processes, and other system-level conditions that may affect performance. The project will be implemented in a Linux environment, preferably Ubuntu, using C/C++ and Linux/POSIX system programming interfaces. The tool will collect system information and monitor resources such as CPU utilization, memory usage, process details, and system status using Linux interfaces such as the /proc filesystem, file I/O, file descriptors, and relevant system calls. The application will periodically read and process system information and present the results in a simple command-line interface. The project demonstrates practical Operating Systems concepts including process management, resource monitoring, file handling, system calls, and interaction with Linux system interfaces. The expected outcome is a functional Linux-based monitoring tool that provides useful real-time or periodic system information while demonstrating the practical application of Operating Systems and Systems Programming concepts.

2. Problem Statement

(Clearly describe the problem that the project intends to address and its relevance to Operating Systems and Systems Programming.)

Linux systems manage various resources such as CPU, memory, processes, and other system components continuously. However, obtaining this information in a simple and organized manner can be difficult for users without directly interacting with Linux system interfaces and commands. There is a need for a lightweight user-space tool that can collect and display important system information and resource usage in a clear format.

The proposed project addresses this problem by developing a Linux System Information and Resource Monitoring Tool using C/C++ and Linux/POSIX system-programming mechanisms. The tool will access Linux system information, particularly through the /proc filesystem and file I/O mechanisms, to obtain details about processes, CPU utilization, memory usage, and overall system status. The project is directly relevant to Operating Systems and Systems Programming because it demonstrates how user-space applications interact with Linux system interfaces and monitor resources managed by the operating system.

The project will provide a functional command-line monitoring tool while demonstrating practical concepts such as process monitoring, file descriptors, file I/O, system calls, and Linux system interfaces.

3. Objectives

(List the specific objectives of the project.)

1. To develop a Linux-based system information and resource monitoring tool using C/C++.
2. To collect and display important system information such as CPU, memory, processes, and system uptime.
3. To monitor system resources such as CPU and memory utilization and present the information in a clear format.
4. To demonstrate Operating Systems and Systems Programming concepts through Linux system interfaces, file handling, system calls, and process management.

4. Proposed Methodology

(Briefly explain how the project will be designed and implemented in Linux. Describe the major functionality, approach, and flow of the proposed solution.)

The project will be designed and implemented as a **user-space Linux application using C/C++**. The tool will collect system information and resource-usage data through **Linux system interfaces and file I/O mechanisms**. The collected data will be processed by the application to calculate and organize CPU, memory, process, and system information. The results will then be displayed through a simple **command-line interface**. The tool will periodically collect updated information to support resource monitoring. The implementation will use relevant **Linux/POSIX APIs, system calls, file descriptors, and process-management concepts** to demonstrate practical Operating Systems and Systems Programming concepts.

5. Operating Systems Concepts / Linux APIs Used

(List the relevant Operating Systems concepts, Linux system calls, POSIX APIs, or mechanisms that will be used.)

OS Concept / Linux API / System Call	Purpose in the Project
System Calls	Interact with Linux kernel services to obtain system information.
File Descriptors	Access and manage system information files.
Process Management	Identify and monitor running processes.
Linux File I/O (`open()`, `read()`, `close()`)	Read system and resource information.
User Space and Kernel Space	Demonstrate how the monitoring tool accesses OS-managed information.
Linux `/proc` Interface	Obtain CPU, memory, process, and system information.

6. Individual Contribution

(Clearly specify the responsibility of each team member in the project.)

Roll Number	Student Name	Individual Responsibility
2520030554	CH. Sai Hima Sekhar	System information module and Linux system interface handling
2520030063	K. Shankar	CPU and memory resource monitoring module
2520030554	V. Navadeep	Process monitoring, testing, and result analysis

7. Tools / Platforms / Software Used

(List the programming languages, operating system, libraries, tools, and software planned for the project.)

Tool / Platform / Software	Purpose
Linux / Ubuntu	Development and execution environment
C / C++	Implement the monitoring tool
GCC Compiler	Compile the C/C++ source code
Linux/POSIX APIs	Access system-level information and OS services
GDB	Automate program compilation and build

8. Expected Outcome

(Describe the expected functionality, output, and outcome of the project.)

The project is expected to produce a **functional Linux-based system information and resource monitoring tool** that displays important system information and monitors resources such as **CPU, memory, and running processes**. The tool will provide the collected information through a simple command-line interface and demonstrate the practical use of **Linux system calls, file I/O, file descriptors, process management, and Linux system interfaces**. The final implementation will help users observe system resource usage while demonstrating the Operating Systems and Systems Programming concepts covered in the course.

9. GitHub Repository

GitHub Repository Link: https://github.com/saihimasekhar-wq/Kernel_Kings_OSSP

Repository Created: ____________________

☐ Yes ☐ No

Faculty Name and Signature

Faculty Name: ____

Signature: ____

Remarks: ____

Faculty Use

Project Approved: ___________________________________

☐ Yes ☐ No

Date: ____

Faculty Signature: ____